Mrunal Hole

Open to relocation within the United States | +1 (614) 419-0147 | hole.17@osu.edu | Portfolio | LinkedIn | GitHub

Summary

Full-stack Software Engineer with 3+ years of experience building scalable backend systems, production APIs, and web applications across Python, TypeScript, Java, and React. Experienced integrating LLM APIs into production systems; led an AI Presidential Challenge team of 8. Passionate about responsible AI and building technology that creates real-world impact.

Education

The Ohio State University — Columbus, OH
M.S. Computer Science and Engineering; GPA: 3.82/4.0 — Aug 2024 – May 2026

Savitribai Phule Pune University — Pune, India
B.E. Information Technology; GPA: 9.30/10 — Aug 2018 – Apr 2022

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, Bash
Backend & Frontend: FastAPI, Spring Boot, REST APIs, GraphQL, Microservices, OOP, System Design, React, Next.js, HTML5, CSS3, Django
Cloud & Infrastructure: AWS (Lambda, S3, RDS, Bedrock), Azure (Synapse, Data Factory), Docker, Kubernetes, CI/CD, GitHub Actions, Git, Linux
AI & LLMs: LLM API Integration (OpenAI, Anthropic/Claude, Gemini), LangChain, LiteLLM, RAG Pipelines, PyTorch, Hugging Face, ML Model Evaluation, Prompt Engineering, Responsible AI
Databases: PostgreSQL, MySQL, MongoDB, Kafka, PySpark, Snowflake, Databricks, Redis

Work Experience

Full Stack Developer Intern, ParaWave: Marysville, OH — May 2025 – Aug 2025
Python, FastAPI, GraphQL, REST APIs, AWS Lambda, AWS Bedrock, Redis, Kafka, Kubernetes, pytest

- Designed, built, and shipped a cost-optimized production-ready RESTful API microservice on AWS Lambda integrating AWS Bedrock, achieving sub-3-second latency and 29% accuracy improvement; implemented OAuth2 authentication, CloudWatch monitoring, and alerting to ensure operational excellence and observability.
- Architected a Redis caching layer over third-party streaming pipelines using caching design patterns, decreasing data retrieval latency by 37%; accelerated prototyping and iterative validation using AI-assisted development tools (GitHub Copilot, Claude Code) without compromising architecture quality.
- Engineered fault-tolerant real-time data ingestion pipelines using Kafka and Kubernetes to surface live conditions including speeds, congestion, and delivery status, reducing video delivery delay to 1–2 seconds under high-concurrency load; validated reliability through A/B testing across distributed streaming architectures.

Senior Software Engineer, Persistent Systems Ltd: Pune, India — Feb 2022 – Jul 2024
Python, JavaScript, PySpark, Databricks, Azure Synapse, Azure Data Factory, DBT, Snowflake, SQL Server, PostgreSQL

- Engineered batch ETL pipelines in Databricks (PySpark) and Azure Synapse using incremental load strategies and CTEs across SQL Server and PostgreSQL, optimizing database schemas and queries to improve pipeline processing efficiency by 29% and reduce data redundancy by 47%.
- Modernized data delivery workflows via Azure Data Factory with API-driven extraction and automated Data Lake integration, accelerating master data management transfer by 34% for thousands of downstream consumers; designed scalable ELT architecture aligned with hybrid cloud processing requirements.
- Collaborated cross-functionally with product and infrastructure teams across 20+ deployments; led code reviews on ETL patterns and CI/CD best practices, reducing deployment time by 28% with zero cross-environment regressions.

Projects

Geospatial Depth Estimation with LLM Evaluation Platform — Aug 2025 – Dec 2025
Python, LangChain, LiteLLM, PyTorch, Hugging Face, PySpark, RAG, LLM API Integration, Rasterio, GeoPandas

- Owned full project lifecycle of a scalable LLM evaluation pipeline integrating third-party APIs via LangChain and RAG, applying responsible AI practices (consistency, accuracy, bias-aware evaluation); engineered data quality controls that reduced manual evaluation effort by 60% across 10,000+ samples.
- Processed 10k+ geospatial records with under 5% data loss by scaling PySpark batch ingestion pipelines to extract and transform spatio-temporal metadata, applying data lake design patterns for PyTorch-based depth prediction modeling.

CV & NLP Model Fine-Tuning and Evaluation — Jan 2025 – May 2025
Python, PyTorch, YOLO, SAM, BERT, Gemini API, Hugging Face, Scikit-learn

- Increased small-object dental pathology detection accuracy by 22% by fine-tuning YOLO with tiled inference (SAHI) and SAM on the AlphaDent Kaggle dataset through multi-scale evaluation and iterative model refinement.
- Reduced NLP classification error rate by 18% by integrating the Gemini API and designing a structured LLM evaluation pipeline with benchmarking across BERT and transformer architectures; documented evaluation methodology as a reusable template, demonstrating awareness of model fairness and reproducibility.

Certifications & Publications

AZ-900: Microsoft Azure Fundamentals | **SnowPro Core COF-CO2**: Snowflake Certification |
Publication: Vehicle Tracking System with Anti-theft Security and Parental Control (May 2022)